

Adem Mrabet

AI Engineer — Agentic Systems & RAG

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PROFESSIONAL SUMMARY

- AI engineer who designs, builds, and ships agentic AI systems end to end: took an open-ended “could AI help here?” problem with no existing dataset or schema and turned it into a production RAG pipeline with its own retrieval, generation, and evaluation loop.
- Built a grounded-generation layer where an LLM only rephrases pre-assembled, fact-checked answers, and an automatic verifier discards any output that drifts from the underlying data — evaluation is a first-class part of the system, not an afterthought.
- Comfortable owning the full stack behind an AI product: hybrid retrieval, orchestration, backend (FastAPI/PostgreSQL), and deployment (Docker), plus a second shipped RAG system (FAISS + BERT embeddings + Groq generation) and a distributed pipeline over large-scale behavioral/transactional data (PySpark/Spark MLlib).

WORK EXPERIENCE

AI Engineer — Final-Year Project (PFE), African Development Bank / Polytech Intl

Apr. 2026 – Sept. 2026 | Tunis, Tunisia

- Took a fuzzy, open-ended AI problem — “can AI make sense of a 250+ page governance document with no existing schema?” — from a vague brief to a shipped, working system, making every call on data model, retrieval strategy, and evaluation criteria.
- Built a deterministic, 8-stage extraction pipeline (character-level PDF parsing with pdfplumber) resolving the source document into a validated knowledge graph of 327 activity nodes, 1,776 responsibility edges, and 131 distinct roles, at a 0.6% unresolved-role rate.
- Designed a hybrid retrieval layer (identifier lookup plus TF-IDF lexical search with deterministic typo correction) feeding a grounded-generation step: an LLM only rephrases pre-assembled, graph-sourced answers, and a verification pass discards any rephrasing that drops a fact — an eval-driven, evaluate-before-ship loop by design.
- Shipped and operated the full system as a production platform — FastAPI, React, PostgreSQL, JWT + Google/Microsoft OAuth, Docker — with real-time conversation sharing (Server-Sent Events), voice input, attachment Q&A, and an analytics dashboard; deployed and publicly reachable.

IT HelpDesk, Outsourcia (on assignment at the African Development Bank)

Since Sept. 2023 | Tunis, Tunisia

- Resolved 1,000+ IT incidents while maintaining business continuity; configured VPN and Remote Desktop access solutions.
- Built PowerBI dashboards and automated Excel reporting for system monitoring, cutting recurring incidents by ~20% and operational risk by 17%.

EDUCATION

Master's Degree in Data Science, Polytech Intl

Since Oct. 2024 | Tunis, Tunisia

Machine Learning | Deep Learning | NLP | Statistics | Optimization | Big Data

Bachelor's Degree in Business Intelligence, Jendouba University

Oct. 2019 – Jun. 2022 | Jendouba, Tunisia

Data Warehousing | Programming (Python, Java, C, C#) | Databases (SQL, NoSQL, PL/SQL) | Data Analysis & Visualization

ACADEMIC PROJECTS

Legal Chatbot — Retrieval-Augmented Generation (RAG) System

Since Dec. 2025 | Polytech Intl

Python, HuggingFace, FAISS, BERT, Groq

- Built a full RAG pipeline to answer questions from multi-PDF legal documents: text extraction, cleaning, chunking, and metadata structuring, followed by BERT embeddings indexed in FAISS for semantic vector search.
- Integrated HuggingFace Transformers with a custom retriever and Groq-based generation to produce structured answers referencing relevant legal articles, in a modular, multi-document-ready architecture.

Credit Card Fraud Detection

Oct. 2025 | Polytech Intl

PySpark, Spark MLlib, Big Data Engineering

- Built a distributed pipeline on PySpark/Spark MLlib (ETL, feature engineering, model training and evaluation across Logistic Regression, Random Forest, and Gradient Boosting) over large-scale transactional data, reaching 73% precision.

CERTIFICATES

Aspire Leadership Program, Aspire Institute (Harvard-affiliated)

Sept. – Dec. 2025 | Cambridge, MA, USA

- Selected finalist among 9,988 participants worldwide (out of 54,337 applicants); completed 40 hours of Harvard-faculty-led coursework on leadership, critical thinking, and AI's societal impact.

Fundamentals of Deep Learning, Nvidia

Oct. 2025 | Tunis, Tunisia

FullStack JS, GoMyCode

Oct. 2022 – May 2023 | Tunis, Tunisia

SKILLS

- **AI / LLM & Agentic Systems:** RAG, agentic pipelines, evaluation-driven development, prompt & context engineering, knowledge graphs, Groq, HuggingFace Transformers, FAISS & vector search, BERT
- **Data & Infra:** SQL, PostgreSQL, Python (NumPy, Pandas, Scikit-learn), PySpark, Spark MLlib, large-scale/behavioral datasets
- **Backend & Deployment:** FastAPI, React, REST APIs, Docker, CI/CD, AWS (S3, Lambda, Athena), JWT / OAuth2, Server-Sent Events
- **Machine Learning & Deep Learning:** CNNs, Transfer Learning, TensorFlow, PyTorch, Keras, Supervised Learning, Feature Engineering, Model Evaluation
- **Other Tools:** Git/GitHub, Jupyter, MLflow, Excel
- **Professional:** Ownership & initiative, fast iteration, clear written communication, adaptability

LANGUAGES

- **Arabic:** Native
- **English:** Fluent
- **French:** Professional working proficiency